
UNIVERSITY OF TORONTO
Computer Science

Winter 2026

Assignment 4
CSC 384 H1

UTORid (e.g. gummalur):

--	--	--	--	--	--	--	--

Student # (9 or 10 digits):

--	--	--	--	--	--	--	--	--	--

First Name:

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last Name:

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Instructors: Chandra Gummaluru, Bahar Aameri

Weight: 5%

Instructions:

- Download this document as a PDF and complete all questions by annotating it. You may either print and write on it, then scan your work, **or** annotate it directly on a tablet/computer.
- Upload your completed PDF to MarkUs.
- Read each question carefully. Marks cannot be awarded for responses based on misinterpretation.
- Show your work where instructed. Answers without clear justification will not receive marks.

Good Luck!

Question 1**[8 pts]****First-Order Logic Translation.**

A caveman is exploring the jungle and encounters various animals and objects. We use the following vocabulary:

Predicates	Meaning
$\text{snake}(x)$	x is a snake
$\text{mushroom}(x)$	x is a mushroom
$\text{poisonous}(x)$	x is poisonous
$\text{eats}(x, y)$	x eats y
$\text{harms}(x, y)$	x harms y
$\text{avoids}(x, y)$	x avoids y
$\text{human}(x)$	x is a human
$\text{warns}(x, y)$	x warns y

Constants: $c0$ (the caveman), $s1$ (a snake), $m1$ (a red mushroom), $m2$ (a brown mushroom).

Translate each of the following English statements into a **sentence** of FOL (no free variables).

- (a) [2 pts] All snakes are poisonous.
- (b) [2 pts] Any poisonous thing harms every human that eats it.
- (c) [2 pts] Every snake avoids everything that is not a mushroom.
- (d) [2 pts] If something warns a human, the human avoids everything that harms it.

Question 2**[10 pts]****Logical Ambiguity via Contradictory Structures.**

Consider the following knowledge base Φ in clausal form:

- (1) $\text{bird}(\mathbf{a})$
- (2) $\text{penguin}(\mathbf{b})$
- (3) $\neg \text{penguin}(x) \vee \text{bird}(x)$
- (4) $\neg \text{penguin}(x) \vee \neg \text{flies}(x)$
- (5) $\neg \text{bird}(x) \vee \text{flies}(x) \vee \text{penguin}(x)$

Consider the query $\Phi' = \text{flies}(\mathbf{a})$.

- (a) [2 pts] Explain briefly (1–2 sentences) why Φ' is *logically ambiguous* given Φ — that is, why it is neither a logical consequence nor a contradiction.

-
- (b) [4 pts] Construct a structure $\mathcal{S}_1$ such that $\mathcal{S}_1 \models \Phi$ **and** $\mathcal{S}_1 \models \Phi'$. Specify the domain $\mathcal{D}$, constant interpretations, and predicate interpretations.
- (c) [4 pts] Construct a structure $\mathcal{S}_2$ such that $\mathcal{S}_2 \models \Phi$ **and** $\mathcal{S}_2 \not\models \Phi'$. Specify the domain $\mathcal{D}$, constant interpretations, and predicate interpretations.

Question 3**[10 pts]****Resolution by Refutation.**

Consider the following knowledge base Φ in clausal form:

- (1) `mammal(c0)`
- (2) `carnivore(c0)`
- (3) `prey(c1)`
- (4) $\neg \text{carnivore}(x) \vee \neg \text{prey}(y) \vee \text{hunts}(x, y)$
- (5) $\neg \text{mammal}(x) \vee \neg \text{hunts}(x, y) \vee \text{tires}(x)$
- (6) $\neg \text{tires}(x) \vee \text{rests}(x)$

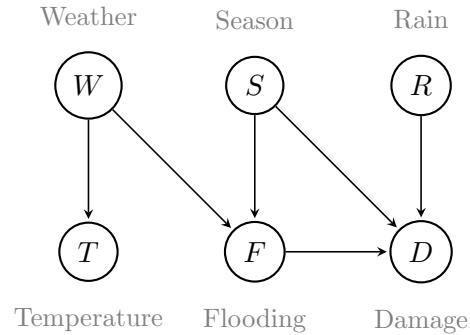
We wish to prove the query $\Phi' = \text{rests}(c0)$ by **resolution by refutation**.

- (a) [2 pts] State the clause that must be added to Φ to begin the proof.
- (b) [8 pts] Carry out the resolution proof. For each step, write the new clause derived and indicate which clauses were resolved and what substitution was used using the notation $r[i, j]\{\delta\}$.

(extra space)

Question 4**[8 pts]****Bayesian Network: D-Separation.**

Consider the Bayesian network below:



For each pair below, state whether the variables are **independent** or **dependent** given the conditioning set $\mathcal{K}$. Justify your answer by identifying the relevant path(s) and junction type(s).

- (a) [2 pts] $W \perp\!\!\!\perp R \mid \emptyset$
- (b) [2 pts] $W \perp\!\!\!\perp R \mid \{F\}$
- (c) [2 pts] $T \perp\!\!\!\perp D \mid \{W, S\}$
- (d) [2 pts] $R \perp\!\!\!\perp W \mid \{D\}$

Question 5**[12 pts]****Bayesian Network: Probabilistic Inference.**

Consider the Bayesian network with three binary variables A , B , C , where $A \rightarrow B$, $A \rightarrow C$, and $B \rightarrow C$ (i.e. C has parents B and A). The conditional probability tables are given below:

A	$P(A)$
0	0.4
1	0.6

A	B	$P(B A)$
0	0	0.7
0	1	0.3
1	0	0.2
1	1	0.8

A	B	C	$P(C A, B)$
0	0	0	0.9
0	0	1	0.1
0	1	0	0.6
0	1	1	0.4
1	0	0	0.5
1	0	1	0.5
1	1	0	0.2
1	1	1	0.8

Compute $P(A = 1 | C = 1)$ **analytically** (using variable elimination). Show all steps, including how you eliminate variables and normalize. [12 pts]

(extra space)

(extra space)